

Unit 1, Lesson 1: Classifying Parent Functions



Benchmark

MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input-output table, determine a function type that could represent it.



Learning Target

- I can classify families of functions from a graph or equation.



1.1.1 Warm-Up: Would You Rather?

- Complete the chart by moving around the room and asking nine different people “Would you rather ...?” Introduce yourself to each person. Ask each person a different question from your chart. Record their name and their answer.

Would you rather ...

fly a kite or swing on a swing?	be a famous inventor or a famous writer?	go skiing or go to a water park?
be able to create a new holiday or create a new sport?	start a colony on another planet or be the leader of a small country on Earth?	own an old pirate ship with crew or a private jet with a pilot?
see a firework display or a circus performance?	be able to talk to land animals or talk to sea animals?	time-travel to the past or the future?

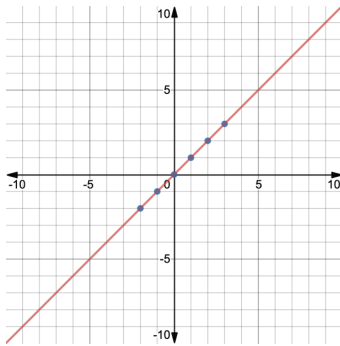


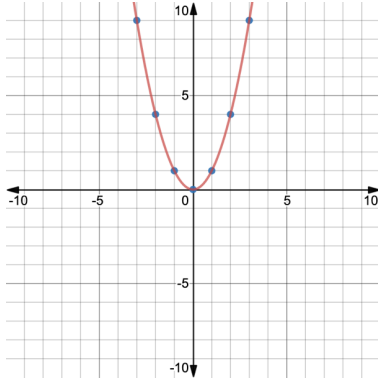


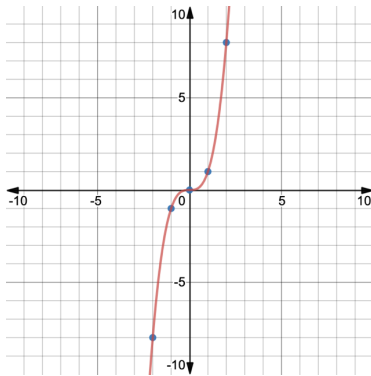
1.1.2 Activity: Parent Functions Gallery Walk

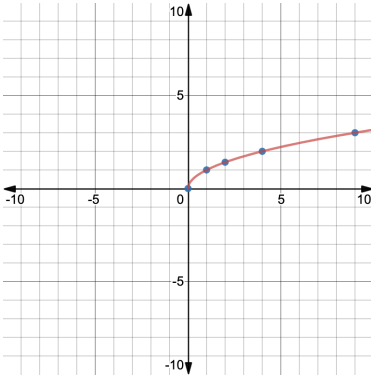
The study of functions is an area of focus in algebra. Functions are classified into families. The **parent function** is the simplest function of each family that preserves the characteristics of the family.

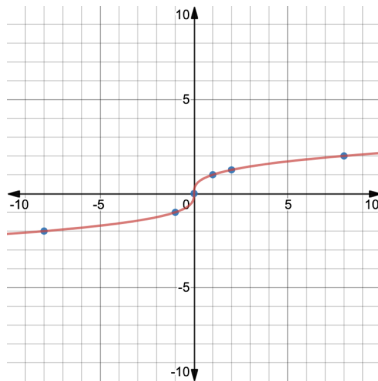
Here are eight parent functions that will be introduced in Algebra 1. After you examine each parent function, answer the questions that follow.

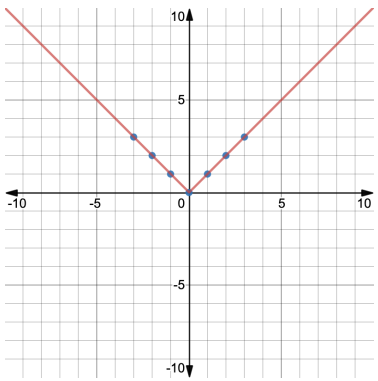
Linear Function																
Parent Function	Table	Graph														
$y = x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table>	x	y	-2	-2	-1	-1	0	0	1	1	2	2	3	3	
x	y															
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3	3															
What do you notice about the graph ?																
What do you notice about the equation ?																

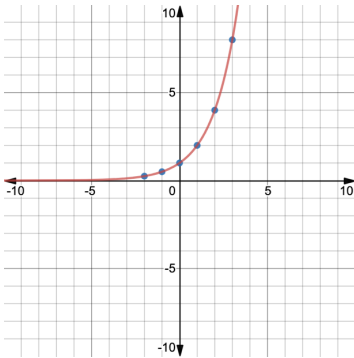
Quadratic Function																
Parent Function	Table	Graph														
$y = x^2$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>9</td></tr></table>	x	y	-2	4	-1	1	0	0	1	1	2	4	3	9	
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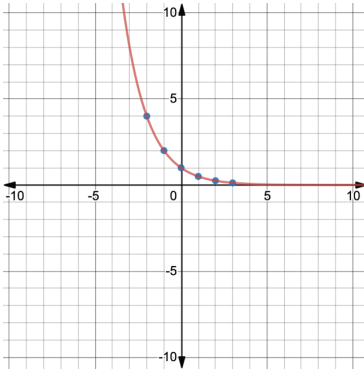
Cubic Function																
Parent Function	Table	Graph														
$y = x^3$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-8</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>8</td></tr><tr><td>3</td><td>27</td></tr></table>	x	y	-2	-8	-1	-1	0	0	1	1	2	8	3	27	
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1	1															
2	8															
3	27															
What do you notice about the graph ?																
What do you notice about the equation ?																

Square Root Function																
Parent Function	Table	Graph														
$y = \sqrt{x}$	<table><tr><th>x</th><th>y</th></tr><tr><td>-1</td><td>not real</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>1.414</td></tr><tr><td>4</td><td>2</td></tr><tr><td>9</td><td>3</td></tr></table>	x	y	-1	not real	0	0	1	1	2	1.414	4	2	9	3	
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Absolute Value Function																
Parent Function	Table	Graph														
$y = x $	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>2</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr></table>	x	y	-2	2	-1	1	0	0	1	1	2	2	3	3	
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3	3															
What do you notice about the graph ?																
What do you notice about the equation ?																

Exponential Function																
Parent Function	Table	Graph														
Exponential Growth Base (2) $y = 2^x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>$\frac{1}{4}$</td></tr><tr><td>-1</td><td>$\frac{1}{2}$</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>8</td></tr></table>	x	y	-2	$\frac{1}{4}$	-1	$\frac{1}{2}$	0	1	1	2	2	4	3	8	
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3	8															
What do you notice about the graph ?																
What do you notice about the equation ?																

Exponential Function																
Parent Function	Table	Graph														
Exponential Decay Base $\left(\frac{1}{2}\right)$ $y = \left(\frac{1}{2}\right)^x$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>2</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>$\frac{1}{2}$</td></tr><tr><td>2</td><td>$\frac{1}{4}$</td></tr><tr><td>3</td><td>$\frac{1}{8}$</td></tr></table>	x	y	-2	4	-1	2	0	1	1	$\frac{1}{2}$	2	$\frac{1}{4}$	3	$\frac{1}{8}$	
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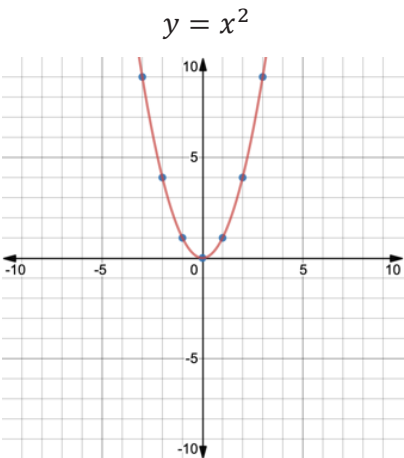


1.1.3 Guided Practice: Classifying Functions

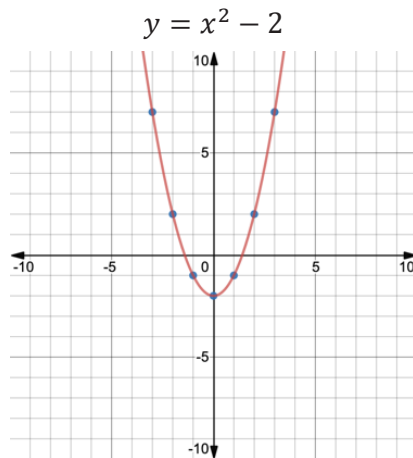
Functions can be transformed by translations, dilations, and reflections.

In this example, the quadratic parent function is translated vertically, horizontally, and reflected over the x -axis.

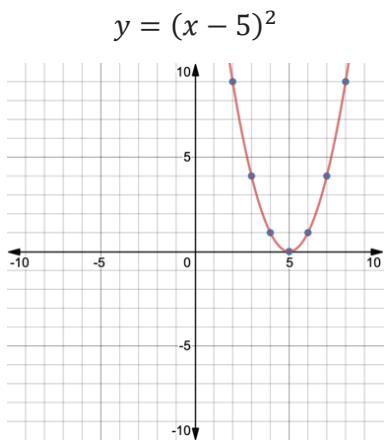
Quadratic Parent Function



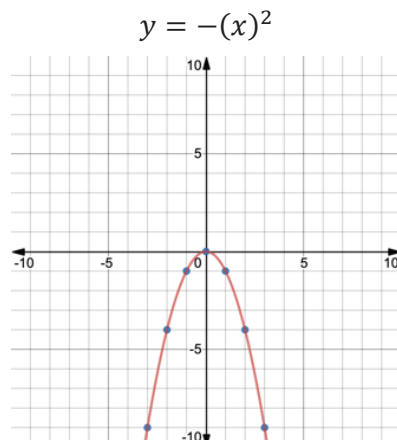
Translated Vertically



Translated Horizontally



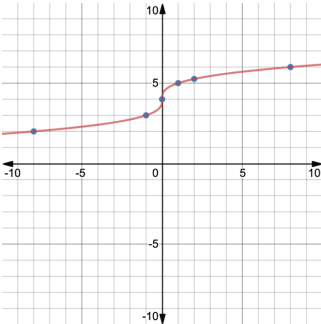
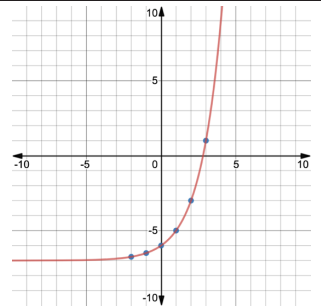
Reflected over the x -axis



1. What stayed the same in each of the equations?

2. What stayed the same in each of the graphs?

3. Classify each function into its function family based on the characteristics observed for the parent functions.

Function	Function Family
$y = x + 9 $	
$y = -\left(\frac{1}{2}\right)^x$	
$y = x^3 - 6$	
	
	



1.1.4 Your Turn!

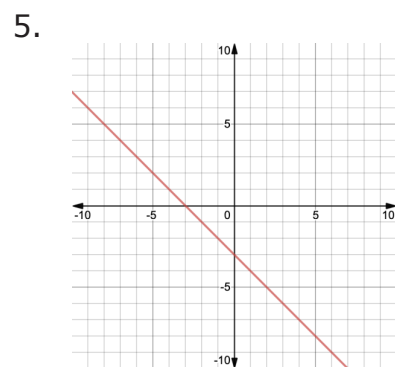
Classify the function family for each equation or graph.

1. $y = (x + 2)^3$ Function Family: _____

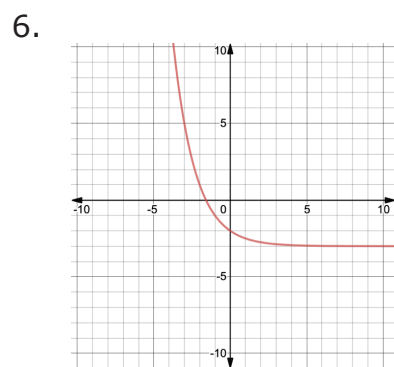
2. $y = -x - 23$ Function Family: _____

3. $y = 2^{(x-7)}$ Function Family: _____

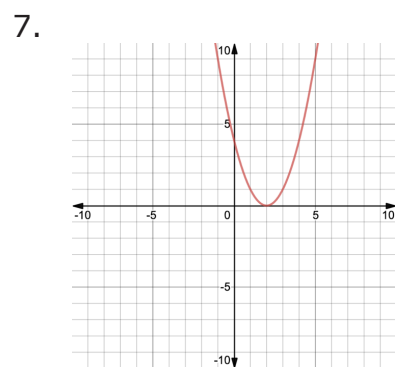
4. $y = (x)^2 + 31$ Function Family: _____



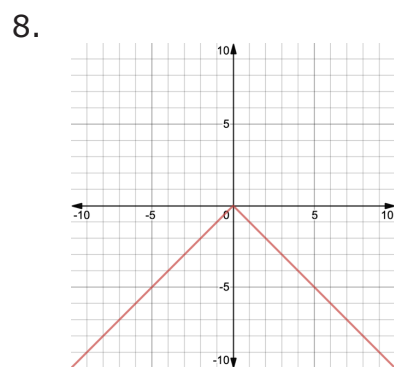
Function Family: _____



Function Family: _____



Function Family: _____



Function Family: _____



1.1.5 Wrap-Up: Reflection

1. As you begin your journey through Algebra 1, rate yourself on where you feel you are today on the following skills.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can evaluate expressions using order of operations.	
I can graph points on a coordinate plane.	
I can apply the laws of exponents.	
I can determine whether a relation is a function.	

2. Rate yourself based on how you feel about the following statements.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am willing to try new things.	
I persevere when something is challenging.	
I am a student learner.	
I am a mathematician.	